

Tian Bai

Ph.D. | UESTC

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SUMMARY

I received my Ph.D. degree in computer science from the University of Electronic Science and Technology of China (UESTC) in June 2024 under the supervision of [Prof. Mingyu Xiao](#). My Ph.D. dissertation is entitled "[Investigations Concerning Parameterized and Exact Algorithms for the Feedback Set and Related Problems](#)". I am an EATCS member and CCF member, and my research interests include graph algorithms, approximation and parameterized algorithms, and algorithmic game theory.

EMPLOYMENT

University of Bergen, Department of Informatics,
Postdoctoral Research Fellow 2025.09 - Present

- Host: Prof. Fedor Fomin.
- Research Interests: Parameterized Algorithms, Optimal Stopping.

The University of Hong Kong, Department of Computer Science,
Postdoctoral Research Fellow 2024.09 - 2025.08

- Host: Prof. Zhiyi Huang.
- Research Interests: Parameterized Algorithms, Contract Design, Prophet Inequality.

EDUCATION

University of Electronic Science and Technology of China, School of Computer Science and Engineering,
Computer Science, *Ph.D.* 2024.06

- Advisor: Prof. Mingyu Xiao.
- Research Interests: Graph Algorithms, Parameterized Algorithms, Mechanism Design.

University of Electronic Science and Technology of China, BS Class (Math & Physics),
Fundamental Sciences (Mathematics and Physics), *B.Sc* 2016.06

- Research Interests: Algebraic Coding Theory, Cryptography.

RESEARCH PUBLICATIONS

Journal Articles

- [1] **Tian Bai**, Mingyu Xiao: [Solving Subset Feedback Vertex Set in Chordal Graphs Faster than \$2^k\$](#) . Information and Computation (Accepted)
- [2] **Tian Bai**, Mingyu Xiao: [A Parameterized Algorithm for Subset Feedback Vertex Set in Tournaments](#). Theoretical Computer Science 975: 114139 (2023).
- [3] **Tian Bai**, Mingyu Xiao: [Exact Algorithms for Restricted Subset Feedback Vertex Set in Chordal and Split Graphs](#). Theoretical Computer Science 984: 114326 (2024).
- [4] **Tian Bai**, Mingyu Xiao: [Computational Complexity of Feedback Vertex Set and Subset Feedback Set Problems: A Survey](#). Journal of computer research and development (2024).
- [5] **Tian Bai**, Mingyu Xiao: Exact Algorithms for Maximum Independent Set Problem on Hypergraphs. SCIENTIA SINICA Informationis.

Conference Proceedings

- [1] **Tian Bai**, Zhiyi Huang, Chui Shan Lee, and Dongchen Li.: [Optimal Stopping with a Predicted Prior](#). Proceedings of the 58th ACM Symposium on Theory of Computing (EC 2026)
- [2] **Tian Bai**, Yiding Feng, Yaohao Liu, Mengfan Ma, and Mingyu Xiao: [Contracting with a Mechanism Designer](#). Proceedings of the 27th ACM Conference on Economics and Computation (EC 2026).
- [3] **Tian Bai**, Yixin Cao, Mingyu Xi: [Feedback Set Problems on \(Planar\) Graphs of Bounded Degrees](#). Proceedings of the 32nd International Computing and Combinatorics Conference (COCOON 2026).
- [4] **Tian Bai**, Fedor Fomin, Petr Golovach, Yash Hiren More, Simon Wietheger Clustering Permutations under the Ulam Metric: A Parameterized Complexity Study Proceedings of the 53rd International Colloquium on Automata, Languages and Programming (ICALP 2026).
- [5] Junqiang Peng, **Tian Bai**, Jingyang Zhao, Mingyu Xiao: [Sustained Vertex Cover on Temporal Graphs](#). Proceedings of the 35th ACM on Web Conference (WWW 2026).

- [6] Yingying Huangfu, **Tian Bai**: [A Cooperative Statistical Approach for Abnormal Node Detection with Adversary Resistance](#). Proceedings of the 34th International Conference on Computer Communications and Networks (ICCCN 2025).
- [7] **Tian Bai**, Mingyu Xiao: [Breaking the Barrier \$2^k\$ for Subset Feedback Vertex Set in Chordal Graphs](#). Proceedings of the 49th International Symposium on Mathematical Foundations of Computer Science (MFCS 2024).
- [8] Mengfan Ma, **Tian Bai**, Mingyu Xiao, Xin Chen: [Facility Assignment with Fair Cost Sharing: Equilibrium and Mechanism Design](#). Proceedings of the 30th International Computing and Combinatorics Conference (COCOON 2024).
- [9] Mengfan Ma, Mingyu Xiao, **Tian Bai**, Bakh Khoussainov: [Facility Location with Entrance Fees](#). Association for the Advancement of Artificial Intelligence (AAAI 2023).
- [10] **Tian Bai**, Mingyu Xiao: [Exact and Parameterized Algorithms for Restricted Subset Feedback Vertex Set in Chordal Graphs](#). Proceedings of the 17th Annual Conference on Theory and Applications of Models of Computation (TAMC 2022).

► TALKS

Presentation: CCF National Conference of TCS (NCTCS) 2025	2025.07
Kernel for Subset Feedback Arc Set in Tournaments.	
Presentation: CCF National Conference of TCS (NCTCS) 2023	2023.07
Computational Complexity of Feedback Vertex Set and Subset Feedback Set Problems: A Survey.	
Presentation: TAMC 2022	2022.09
Exact and Parameterized Algorithms for Restricted Subset Feedback Vertex Set in Chordal Graphs.	

► SERVICES AND PROFESSIONAL ACTIVITIES

Reviewer for Conferences

- ICALP 2026
- ESA 2026
- ISAAC 2024
- MFCS 2024
- IPEC 2025
- COCOON 2025
- SOFSEM 2025
- WALCOM 2022 2024 2025 2026.

Reviewer for Journals

- Theoretical Computer Science.
- Journal of Computer and System Sciences.
- Discrete Applied Mathematics.
- Frontiers of Computer Science.

Moderator for Workshops

- The 19th International Conference and Workshops on Algorithms and Computation (WALCOM), 2025.
- CCF National Conference of Computational Economics, 2024.
- Algorithms & Logic Workshop at UESTC, 2023.
- Algorithmic Game Theory and Computation Theory Forum at UESTC, 2022.
- The 2nd Theoretical Computer Science Outstanding PhD Students Forum (TCSPhd2021), 2021.

►TEACHING

Teaching Assistant for graduate students: Frontier Topics in Doctoral Research	2022-2023
Teaching Assistant for graduate students: Advanced Algorithm Analysis and Design	2022-2023
Teaching Assistant for undergraduate students: Frontiers of Theory of Computation	2022-2023
Teaching Assistant for undergraduate students: Frontiers of Theory of Computation	2021-2022
Teaching Assistant for undergraduate students: Advanced Algebra	2017-2018
Teaching Assistant for undergraduate students: Linear Algebra	2016-2017

►AWARDS AND HONORS

Outstanding Teaching Assistant	2024
Global 69th/China 10th place in IEEE Global Extreme Programming Competition 12.0	2018
First prize in the National University Mathematics Competition	2014 & 2015